EEE407 Tutorial 4a solutions

1. According to MS 1525 (in compliance with MS ISO 8995 and MS 825), what is the maximum number of 36 W fluorescent tubes that could be installed in an office area of $100m^2$.

2. A small light source with intensity uniform in all directions is mounted at a height of 10 metres above a horizontal surface. Two points A and B both lie on the surface with point A directly beneath the source. How far is B from A if the illumination at B is only 1/10 as great as at A?

3. A corridor is lighted by 4 lamps spaced 10 m apart and suspended at a height of 5 m above the centre line of the floor. If each lamp gives a luminous intensity of 200 Cd in all directions below the horizontal, find the illumination at the point on the floor mid-way between the second and third lamps.

4. Two lamps A and B of 200 Cd and 400 Cd, respectively are situated 100 m apart. The height of A above the ground level is 10 m and that of B is 20 m. If a photometer is placed at the centre of the line joining the two lamp posts, calculate its reading.

5. The average luminous output of an 80-W fluorescent lamp 1.5 metre in length and 3.5 cm diameter is 3300 lumens.
   (a) Calculate its average luminance or brightness (cd/m^2).
   (b) If the auxiliary gear associated with the lamp consumes a load equivalent to 25 percent of the lamp, calculate the cost of running a twin unit for 2500 hours at RM0.334/ kWh.

6. After a careful calculation, you have estimated a minimum of 5000 lumens is needed to light up a specific space in your house. The space was previously fitted with several incandescent bulbs using e27 holder. Hence you are looking to retrofit the existing fittings with newer bulbs. Compare these 2 types of light bulbs and calculate life cycle cost of 30,000 hours of usage given the information below:

	Philips-tornado-spiral	Philips-corepro-led
Power rating	23 W	10.5 W
Lumens	1400	1055
Cost	$ 7.51	$ 3.14
Lifespan	10,000 hr	15,000 hr

7. A library has an area of 35m × 25m and the utilization factor of the interiors is 0.95. The lux requirement is 500 lux. You are given 2 choices of light tubes:
   - Fluorescent tube, 6300 lm, 80 W with maintenance factor of 0.75, cost per tube: $18
   - LED tube, 1600 lm, 16 W with maintenance factor of 0.95, cost per tube: $13
     a. Determine the number of fluorescent and LED light tubes required according to the lux requirement.
     b. Determine the energy consumption and lighting power intensity for the 2 light tubes based on number of light tubes required for maximum lux for a year, assuming that this library operates 9 hours a day.